Crypto Pump-and-Dump Detection

Group 2

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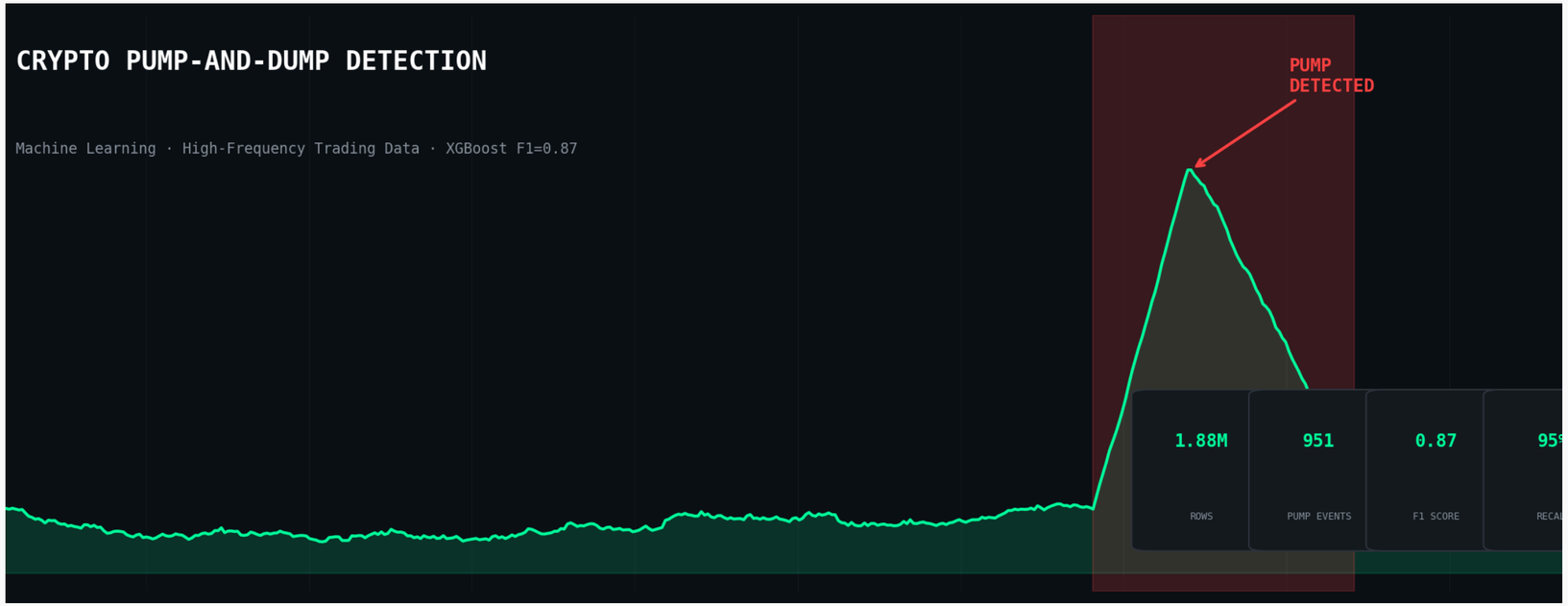

Problem Statement

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The Problem

- Crypto markets are vulnerable to coordinated pump-and-dump schemes
- Artificial price inflation followed by rapid sell-offs
- Leads to financial losses and market distortion
- Detection is challenging due to extreme class imbalance

Project Objective:

Objective

Primary Goal:

- Develop a machine learning model to detect pump events using short time-window trading features

Secondary Goals:

- Handle extreme class imbalance
- Compare multiple models
- Use time-based validation
- Evaluate using precision-recall metrics

.

Dataset Overview

Dataset Description

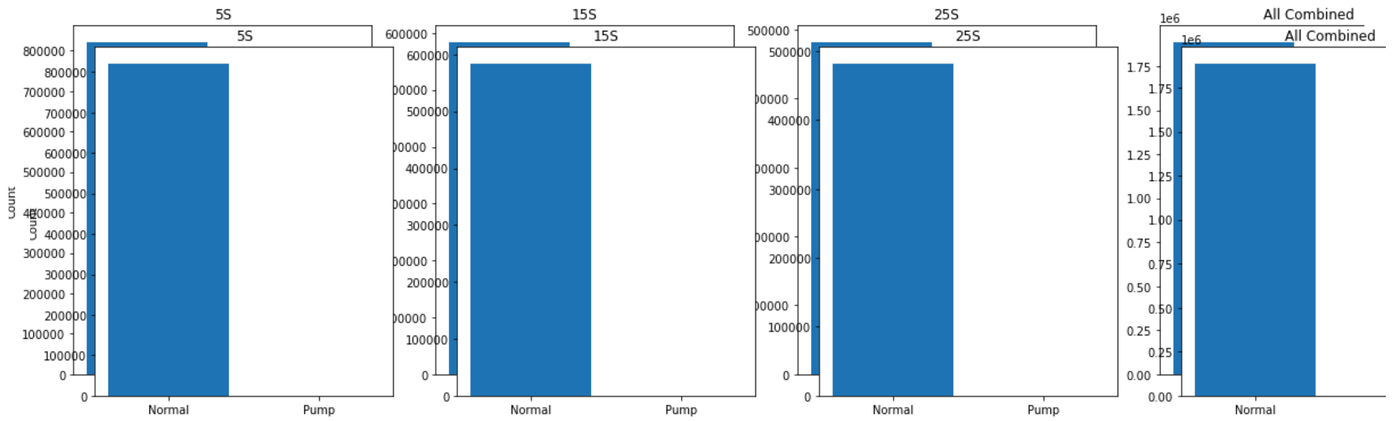
High-frequency crypto market data at:

- 5-second intervals
- 15-second intervals
- 25-second intervals

Target Variable:

- gt = 1 → Pump event
- gt = 0 → Normal activity

Pump events represent only 0.0657% of the data.



Why Multiple Time Windows?

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Why 5s, 15s, 25s?

- 5s → Detects micro-spikes (early signal)
- 15s → Captures short bursts more reliably
- 25s → Detects sustained abnormal activity

Insight:

5s detects early spikes → 15s confirms escalation → 25s confirms trend.

Feature Engineering

Feature Engineering

Key Feature Categories

Volume & Activity

- std_volume
- avg_volume
- std_trades

Price Dynamics

- std_price
- avg_price
- avg_price_max

Order Intensity

- std_rush_order
- avg_rush_order

Exploratory Data Analysis

Exploratory Data Analysis

Pump vs Normal Behavior (5S Data)

Pump events show:

- Much higher std_rush_order
- Significant volume spikes
- Increased trade variability

Example:

std_rush_order during pump ≈ 50–100× higher than normal

Time-Based Validation

Why Not Random Split?

- Data is time-series
- Random split causes temporal leakage
- Model could “learn from the future”

Solution:

- Train on data before May 2020
- Test on data after May 2020

Modelling

Models Evaluated

1. Logistic Regression
2. Random Forest
3. XGBoost

Evaluation Focus:

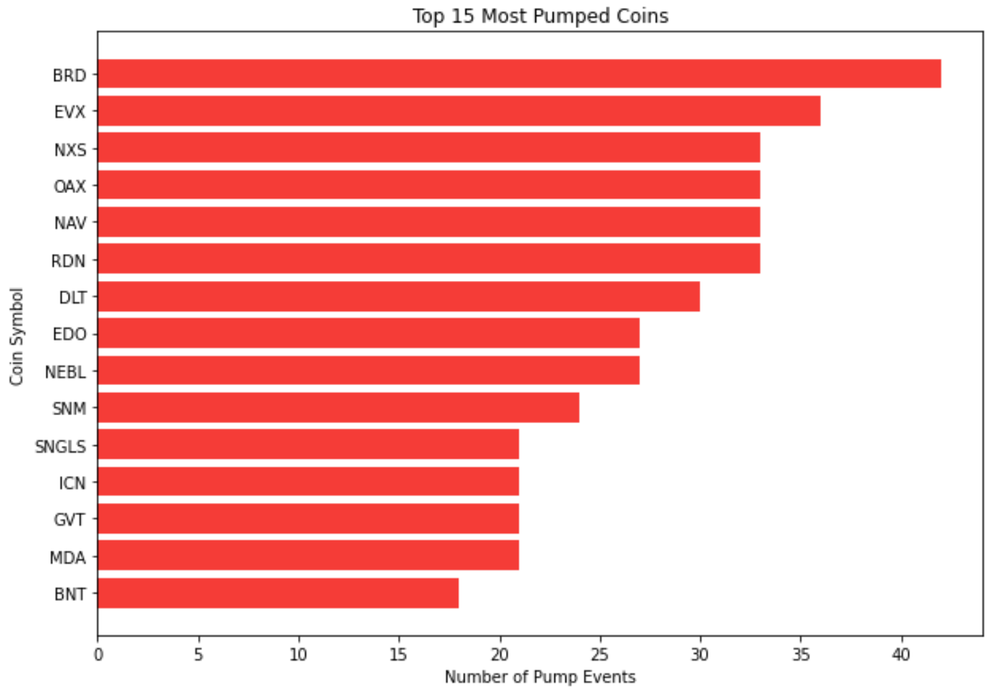
- Precision (Class 1)
- Recall (Class 1)
- F1-Score
- Precision-Recall tradeoff

Results.

Model	Precision	Recall	F1-Score
Logistic Reg.	0.12	1	0.22
Random Forest	0.71	0.95	0.82
XGBoost	0.79	0.95	0.87

Key Findings

- XGBoost is highly effective for rare-event detection
- Volume and order intensity are strong pump indicators
- Certain tokens appear more vulnerable to manipulation
- Accuracy is misleading for imbalanced datasets



Business Impact and Conclusion

- Deploy real-time ML monitoring for pump detection.
- Avoid high-risk, low-liquidity tokens with sudden price surges.
- Maintain ongoing model retraining and surveillance updates.

Thank You For Your Time